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Analytical T-Matrix for Elastic Scatterers: Sphere and Cube Geometries — Derived from the Lippmann–Schwinger Integral Equation

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SUMMARY

We apply symbolic computer algebra (SymPy) to evaluate the Lippmann–Schwinger integral equation analytically for two scatterer geometries: a sphere and a cube, both with constant isotropic contrasts $\Delta\lambda$, $\Delta\mu$, $\Delta\rho$, embedded in a homogeneous isotropic background. The sphere yields closed-form results with isotropic effective contrasts; the cube introduces cubic anisotropy in the effective stiffness, providing the natural building block for effective medium theories of heterogeneous solids. Self-consistent amplification factors are derived that sum the multiple-scattering series inside each inclusion exactly, yielding a T-matrix that satisfies the optical theorem. The local site T-matrix is then embedded in the Foldy–Lax multiple-scattering framework with the inter-site Green's tensor provided by the Kennett reflectivity propagator for layered media. A hybrid numerical implementation in NumPy achieves $\sim 10^{-8}$ relative accuracy by combining Gauss–Legendre quadrature for the convergent Green's tensor integral with a polynomial Taylor expansion that avoids the divergent Eshelby singularity in the second-derivative integrals. The displacement-traction 6×6 T-matrix is derived for direct integration into the Kennett propagator framework.

Key words: Seismic wave propagation – Computational seismology – Wave scattering and diffraction – Numerical modelling – Theoretical seismology.

1 INTRODUCTION

The scattering of elastic waves by localised heterogeneities is a fundamental problem in seismology, ultrasonics, and geophysical exploration. When the scatterer is a sphere of radius a embedded in a homogeneous isotropic background, the problem admits an exact solution in terms of partial-wave (Mie) series (Korneev & Johnson 1993; Gritto et al. 1995). In the long-wavelength (Rayleigh) regime $ka \ll 1$, only the lowest-order multipoles contribute and the scattered field can be expressed compactly through an effective T-matrix that maps the incident wavefield to the scattered wavefield.

The T-matrix concept originates in quantum scattering theory and was adapted to elastic-wave problems by Waterman (1976); Bostock (1991) and Bostock & Kennett (1992) applied it to surface-wave scattering. Pollitz (1998) derived the scattered field for a small spherical inclusion without assuming small perturbations, arriving at self-consistent amplification factors that are functionally equivalent to renormalised material contrasts. Pellegrini et al. (2011) formalised this as a mean-field T-matrix approach: by approximating the interior stress and momentum fields by their volume averages, the full multiple-scattering series inside the inclusion can be summed in closed form, yielding a T-matrix that satisfies the optical theorem exactly.

On the computational side, Jakobsen and co-workers developed integral-equation methods for seismic modelling and inversion based on the Lippmann–Schwinger equation. Jakobsen (2012) introduced the T-matrix formalism for acoustic forward modelling; Jakobsen & Ursin (2015) extended this to direct iterative T-matrix inversion; and Jakobsen & Wu (2016) proposed a renormalised scattering series for frequency-domain waveform modelling in the presence of strong velocity contrasts, overcoming the convergence limitations of the Born series. Most recently, Shekhar et al. (2024) formulated a numerical integral-equation method for microseismic wavefield modelling in arbitrarily anisotropic elastic media, providing the Green's tensor expressions and the Lippmann–Schwinger framework used as the starting point of this work.

In this note, we apply symbolic computer algebra (SymPy) to evaluate the Lippmann–Schwinger integral equation analytically for two scatterer geometries: a sphere and a cube, both with constant isotropic contrasts $\Delta\lambda$, $\Delta\mu$, $\Delta\rho$. The sphere yields closed-form results with isotropic effective contrasts; the cube introduces cubic anisotropy in the effective stiffness, providing the natural building block for effective medium theories of heterogeneous solids.

Part I: Spherical Scatterer

2 PROBLEM STATEMENT

We consider the scattering of elastic waves by a small spherical inclusion of radius a embedded in an infinite homogeneous isotropic background medium with P-wave velocity α, S-wave velocity β, and density ρ. The inclusion has constant isotropic contrasts $\Delta\lambda$, $\Delta\mu$, $\Delta\rho$ relative to the background.

The wavefield satisfies the Lippmann–Schwinger integral equation (Eq. 8 of Shekhar2024):

$$u_i(\mathbf{x}) = u_i^0(\mathbf{x}) + \int_D \Big[G_{ij}(\mathbf{x},\mathbf{x}')\,\omega^2\,\Delta\rho\; u_j(\mathbf{x}') + H_{ijk}(\mathbf{x},\mathbf{x}')\,\Delta c_{jklm}\,\varepsilon_{lm}(\mathbf{x}') \Big] d^3x', \tag{1}$$

where the strain–Green's tensor is

$$H_{ijk}^{(0)}(\mathbf{x},\mathbf{x}') = \tfrac{1}{2}\big(G_{ij,k} + G_{ik,j}\big), \tag{2}$$

and the isotropic stiffness contrast is

$$\Delta c_{jklm} = \Delta\lambda\,\delta_{jk}\,\delta_{lm} + \Delta\mu\,(\delta_{jl}\,\delta_{km} + \delta_{jm}\,\delta_{kl}). \tag{3}$$

3 BACKGROUND GREEN'S TENSOR

The elastodynamic Green's tensor for a homogeneous isotropic medium (Appendix A of Shekhar2024) admits the radial decomposition:

$$G_{ij}(\mathbf{x}) = f(r)\,\delta_{ij} + g(r)\,\hat{n}_i\,\hat{n}_j, \tag{4}$$

with $\hat{n}_i = x_i/r$ and radial functions

$$X(r) = \frac{e^{i\omega r/\alpha}}{4\pi\rho\alpha^2}, \quad V(r) = \frac{e^{i\omega r/\beta}}{4\pi\rho\beta^2}, \quad C = \frac{1-\beta^2/\alpha^2}{8\pi\rho\beta^2}, \qquad f(r) = \frac{V(r)}{r} - \frac{C}{r}, \tag{5}$$

$$g(r) = \frac{3C}{r} - \frac{X(r)}{r} + \frac{V(r)}{r}. \tag{6}$$

4 CALCULUS: SPHERICAL COORDINATE REDUCTION

The central computational challenge is to evaluate volume integrals of the form $\int_{\text{sphere}} G_{ij}\,d^3x$, $\int_{\text{sphere}} G_{ij,k}\,d^3x$ and $\int_{\text{sphere}} G_{ij,kl}\,d^3x$ over a sphere of radius a centred at the scatterer. The key insight is that although these integrals live in Cartesian tensor space, the radial decomposition (4) allows them to be factored into angular integrals (which evaluate to known isotropic tensor identities) multiplied by 1D radial integrals (which SymPy evaluates in closed form).

4.1 Conversion to Spherical Coordinates

We write the volume element as $d^3x = r^2\, dr\, d\Omega$, where $d\Omega = \sin\theta\, d\theta\, d\varphi$ is the solid-angle element. Every Cartesian integrand is expressed in terms of the radial coordinate r and the unit direction vector $\hat{n}_i = x_i/r$. Because $f(r)$ and $g(r)$ depend only on r, the angular and radial parts separate cleanly:

$$\int_{\text{sphere}} \big[f(r)\,\delta_{ij} + g(r)\,\hat{n}_i\,\hat{n}_j\big] r^2\, dr\, d\Omega = \delta_{ij} \int_0^a r^2 f(r)\, dr \int d\Omega + \int_0^a r^2 g(r)\, dr \int d\Omega\; \hat{n}_i \hat{n}_j. \quad (7)$$

4.2 Angular Integral Identities

The angular integrals of products of direction cosines over the unit sphere are determined entirely by symmetry. The only non-vanishing cases with an even number of $\hat{n}$ factors are:

$$\int d\Omega = 4\pi, \quad (8)$$

$$\int d\Omega\; \hat{n}_i\,\hat{n}_j = \frac{4\pi}{3}\,\delta_{ij}, \quad (9)$$

$$\int d\Omega\; \hat{n}_i\,\hat{n}_j\,\hat{n}_k\,\hat{n}_l = \frac{4\pi}{15}\,(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}). \quad (10)$$

Integrals with an odd number of $\hat{n}$ factors vanish by parity. This is the reason that $\int H_{ijk}\, d^3x = 0$: each term in $G_{ij,k}$ contains an odd number of direction cosines after the chain rule is applied, so the angular integral is zero.

4.3 Chain-Rule Decomposition of Derivatives

The Cartesian derivatives of G_{ij} must be expressed in terms of $\hat{n}_i$ and radial derivatives of f and g before integration. We use the fundamental identities:

$$\frac{\partial r}{\partial x_k} = \hat{n}_k, \qquad \frac{\partial \hat{n}_i}{\partial x_k} = \frac{1}{r}\big(\delta_{ik} - \hat{n}_i\,\hat{n}_k\big). \quad (11)$$

First derivative. Applying the chain rule to $G_{ij} = f\,\delta_{ij} + g\,\hat{n}_i\hat{n}_j$:

$$G_{ij,k} = f'\,\delta_{ij}\,\hat{n}_k + g'\,\hat{n}_i\,\hat{n}_j\,\hat{n}_k + \frac{g}{r}\big(\delta_{ik}\,\hat{n}_j + \delta_{jk}\,\hat{n}_i - 2\hat{n}_i\,\hat{n}_j\,\hat{n}_k\big), \quad (12)$$

where primes denote d/dr. Every term contains an odd number of $\hat{n}$ factors (one or three), confirming that $\int G_{ij,k}\, d^3x = 0$.

Second derivative. Differentiating (12) with respect to x_l and again applying (11) produces terms with angular structures involving 0, 2, or 4 direction cosines:

$$G_{ij,kl} = \text{terms proportional to } \delta\delta,\ \delta\hat{n}\hat{n},\ \hat{n}\hat{n}\hat{n}\hat{n}, \quad (13)$$

with coefficients built from f, g and their first and second radial derivatives. Rather than writing out

all terms explicitly, we use a *contraction strategy*: the full tensor $\int G_{ij,kl}\, d^3x$ must have the isotropic form $A\,\delta_{ij}\delta_{kl} + B\,(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$, and we can determine the two unknowns A and B from any two independent contractions.

4.4 Contraction Strategy for A and B

Contraction 1: Trace over (i, j). Setting $i = j$ and summing gives $G_{ii} = 3f + g \equiv h(r)$, which is a scalar function of r. Its second Cartesian derivative is simply:

$$G_{ii,kl} = h''(r)\,\hat{n}_k\,\hat{n}_l + \frac{h'(r)}{r}\big(\delta_{kl} - \hat{n}_k\,\hat{n}_l\big). \tag{14}$$

Integrating over the sphere using (8)–(9):

$$\int G_{ii,kl}\, d^3x = \delta_{kl} \int_0^a r^2 \left[\frac{4\pi}{3}\, h'' + \frac{8\pi}{3}\,\frac{h'}{r}\right] dr = (3A + 2B)\,\delta_{kl}. \tag{15}$$

Contraction 2: Divergence structure. Summing over $j = k$ gives the divergence $G_{ij,j} = q(r)\,\hat{n}_i$, where $q(r) = f' + g' + 2g/r$. Its further derivative with respect to x_l has the same scalar-times-vector structure:

$$G_{ij,jl} = q'(r)\,\hat{n}_i\,\hat{n}_l + \frac{q(r)}{r}\big(\delta_{il} - \hat{n}_i\,\hat{n}_l\big). \tag{16}$$

Integrating:

$$\int G_{ij,jl}\, d^3x = \delta_{il} \int_0^a r^2 \left[\frac{4\pi}{3}\, q' + \frac{8\pi}{3}\,\frac{q}{r}\right] dr = (A + 4B)\,\delta_{il}. \tag{17}$$

Solving the 2×2 system (15)–(17) yields A and B, with only 1D radial integrals for SymPy to evaluate.

5 ANALYTICAL VOLUME INTEGRALS

5.1 Green's Tensor Integral

Using the decomposition of Section 4, the volume integral of the Green's tensor reduces to:

$$\int_{\text{sphere}} G_{ij}\, d^3x = \delta_{ij} \int_0^a r^2 \left[4\pi\, f(r) + \frac{4\pi}{3}\, g(r)\right] dr \equiv \Gamma_0\,\delta_{ij}. \tag{18}$$

After exact radial evaluation by SymPy:

$$\Gamma_0 = \frac{-3\alpha\beta + 2\alpha\big(\beta - i\omega a\big)\, e^{i\omega a/\beta} + \beta\big(\alpha - i\omega a\big)\, e^{i\omega a/\alpha}}{3\alpha\beta\,\omega^2\rho}. \tag{19}$$

5.2 Second-Derivative Integral

As derived in Section 4.4, the volume integral of $G_{ij,kl}$ has the isotropic structure

$$\int_{\text{sphere}} G_{ij,kl}(\mathbf{x})\, d^3x = A\,\delta_{ij}\delta_{kl} + B\,(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}), \tag{20}$$

with A and B determined by the two contractions (15)–(17). Defining $\xi_P = \omega a/\alpha$ and $\xi_S = \omega a/\beta$, the exact closed-form results are:

$$A = \frac{2\alpha^3\beta\big(1-e^{i\xi_S}\big) + 4i\alpha^3\omega a\, e^{i\xi_S} + 3\alpha\beta^3\big(1-e^{i\xi_P}\big) + i\beta^3\omega a\, e^{i\xi_P}}{15\alpha^3\beta^3\rho}, \tag{21}$$

$$B = \frac{2\alpha^3\beta\big(1-e^{i\xi_S}\big) - i\alpha^3\omega a\, e^{i\xi_S} - 2\alpha\beta^3\big(1-e^{i\xi_P}\big) + i\beta^3\omega a\, e^{i\xi_P}}{15\alpha^3\beta^3\rho}. \tag{22}$$

Note: The parity argument shows $\int H_{ijk}\, d^3x = 0$, so the stiffness contrast enters only through the strain equation.

6 T-MATRIX COUPLING COEFFICIENTS

The T-matrix tensor, defined by contracting the symmetrised second-derivative integral with the stiffness contrast, is isotropic:

$$T_{mnlp} = T_1\, \delta_{mn}\delta_{lp} + T_2\, (\delta_{ml}\delta_{np} + \delta_{mp}\delta_{nl}), \tag{23}$$

where:

$$\begin{aligned} T_1 = \frac{1}{15\alpha^3\beta^3\rho}\Big[&\Delta\lambda\Big(10\alpha^3\beta\big(1-e^{i\xi_S}\big) + 5\alpha\beta^3\big(e^{i\xi_P}-1\big) + 5i\beta^3\omega a\, e^{i\xi_S}\Big) \\ &+ \Delta\mu\Big(4\alpha^3\beta\big(1-e^{i\xi_S}\big) - 2i\alpha^3\omega a\, e^{i\xi_S} + 4\alpha\beta^3\big(e^{i\xi_P}-1\big) + 2i\beta^3\omega a\, e^{i\xi_P}\Big)\Big], \end{aligned} \tag{24}$$

with $\xi_P = \omega a/\alpha$ and $\xi_S = \omega a/\beta$.

$$T_2 = \frac{\Delta\mu}{15\alpha^3\beta^3\rho}\Big[4\alpha^3\beta\big(1-e^{i\xi_S}\big) + 3i\alpha^3\omega a\, e^{i\xi_S} + \alpha\beta^3\big(1-e^{i\xi_P}\big) + 2i\beta^3\omega a\, e^{i\xi_P}\Big]. \tag{25}$$

7 SELF-CONSISTENT SOLUTION

Under the assumption that the wavefield is approximately constant inside the small sphere, the Lippmann–Schwinger equation reduces to a self-consistent algebraic system that decouples into three independent scalar equations for displacement, dilatation, and deviatoric strain, with amplification factors:

$$A_u = \frac{1}{1-\omega^2\Delta\rho\,\Gamma_0}, \qquad A_\theta = \frac{1}{1-3T_1-2T_2}, \qquad A_e = \frac{1}{1-2T_2}. \tag{26}$$

8 EFFECTIVE (RENORMALISED) CONTRASTS — THE T-MATRIX

The scattered field takes the Born form with renormalised contrasts:

$$u_i^{\text{scat}}(\mathbf{x}) = V\Big[H_{ijk}(\mathbf{x},\mathbf{x}_s)\,\big(\Delta\lambda^*\,\theta^0\,\delta_{jk} + 2\Delta\mu^*\,\varepsilon^0_{jk}\big) + \omega^2\, G_{ij}(\mathbf{x},\mathbf{x}_s)\,\Delta\rho^*\,u^0_j\Big], \tag{27}$$

where $V = \frac{4}{3}\pi a^3$ is the sphere volume. The effective contrasts are:

Density:

$$\Delta\rho^* = \frac{\Delta\rho}{1 - \omega^2 \Delta\rho\, \Gamma_0}. \tag{28}$$

Shear modulus:

$$\Delta\mu^* = \frac{\Delta\mu}{1 - 2T_2}. \tag{29}$$

Lamé parameter:

$$\Delta\lambda^* = \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{1 - 3T_1 - 2T_2} - \frac{\frac{2}{3}\Delta\mu}{1 - 2T_2}. \tag{30}$$

9 BORN APPROXIMATION VERIFICATION

In the Born (point-scatterer) limit $a \to 0$, all amplification factors tend to unity:

$$\lim_{a\to 0} \Delta\rho^* = \Delta\rho, \qquad \lim_{a\to 0} \Delta\lambda^* = \Delta\lambda, \qquad \lim_{a\to 0} \Delta\mu^* = \Delta\mu. \tag{31}$$

All three limits have been verified analytically with SymPy.

10 NUMERICAL EXAMPLE

Background: $V_P = 5000$ m/s, $V_S = 3000$ m/s, $\rho = 2500$ kg/m^3. Contrasts: $\Delta\lambda = 2$ GPa, $\Delta\mu = 1$ GPa, $\Delta\rho = 100$ kg/m^3. Frequency $f = 10$ Hz, sphere radius $a = 10$ m.

Quantity	**Value**
$k_P a$	0.1257
$k_S a$	0.2094
Γ_0	$1.730890\text{e}{-09} + 2.282529\text{e}{-10}\, i$
A	$-3.781591\text{e}{-13} + 9.278517\text{e}{-13}\, i$
B	$2.248959\text{e}{-13} - 1.438708\text{e}{-12}\, i$
T_1	$1.492641\text{e}{-03} - 1.253138\text{e}{-02}\, i$
T_2	$-1.532633\text{e}{-04} - 5.108564\text{e}{-04}\, i$
A_u	$1.000684\text{e}{+00} + 9.023389\text{e}{-05}\, i$
A_θ	$1.002681\text{e}{+00} - 3.888157\text{e}{-02}\, i$
A_e	$9.996925\text{e}{-01} - 1.021086\text{e}{-03}\, i$
$\Delta\rho^*$	$1.000684\text{e}{+02} + 9.023389\text{e}{-03}\, i$
$\Delta\lambda^*$	$2.007355\text{e}{+09} - 1.030035\text{e}{+08}\, i$
$\Delta\mu^*$	$9.996925\text{e}{+08} - 1.021086\text{e}{+06}\, i$

Part II: Cubic Scatterer

11 MOTIVATION: WHY CUBES?

The T-matrix for a cube is fundamentally more useful for effective medium theory than the T-matrix for a sphere, because cubes tile three-dimensional space.

A sphere is the natural scatterer for studying isolated inclusions in a matrix, but when the goal is to model a heterogeneous medium as a homogenised effective material, one must partition the entire volume into non-overlapping cells. Spheres cannot fill space without gaps: even the densest packing of equal spheres leaves roughly 26% void (Hales 2005). Cubes, by contrast, are space-filling: they tesselate $\mathbb{R}^3$ with zero overlap and zero void.

This simple geometric fact has far-reaching consequences for effective medium theory (see Section 17).

11.1 Semi-Analytical Approach

For the sphere, the radial decomposition of G_{ij} and the angular integral identities allow exact closed-form evaluation of all volume integrals. For the cube, no such radial-angular factorisation exists. Instead, we use a semi-analytical approach:

(i) Taylor-expand the Green's tensor $G_{ij}(\mathbf{x})$ about the centre of the cube ($\mathbf{x} = \mathbf{0}$) as a polynomial in (x_1, x_2, x_3), keeping terms through order $(\omega a)^7$.

(ii) Integrate each monomial $x_1^{p_1} x_2^{p_2} x_3^{p_3}$ over the cube $[-a, a]^3$ analytically using the factorisation:

$$\int_{-a}^{a}\int_{-a}^{a}\int_{-a}^{a} x_1^{p_1} x_2^{p_2} x_3^{p_3}\, d^3x = \prod_{m=1}^{3} \frac{2\, a^{p_m+1}}{p_m+1} \qquad \text{(all } p_m \text{ even; zero otherwise).} \tag{32}$$

This gives the same leading-order behaviour as the sphere (the $O(ka)^3$ Rayleigh terms), plus a cubic-anisotropy correction at $O(ka)^7$ that has no analogue for the sphere.

12 CUBIC ANISOTROPY TENSOR

The critical difference between sphere and cube appears in the fourth moment of the scatterer volume. For any centrosymmetric shape:

$$\int_D x_i\, x_j\, x_k\, x_l\, d^3x = P\,(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + Q\, E_{ijkl}, \tag{33}$$

where $E_{ijkl} = \sum_m \delta_{im}\delta_{jm}\delta_{km}\delta_{lm}$ is the *cubic anisotropy tensor* (equals 1 when all four indices are equal, zero otherwise).

Sphere: By spherical symmetry, $Q = 0$. The fourth moment is purely isotropic.

Cube $[-a, a]^3$: Direct computation gives

$$\int x_1^4 \, d^3x = \frac{8a^7}{5}, \qquad \int x_1^2 x_2^2 \, d^3x = \frac{8a^7}{9}, \tag{34}$$

so $P = 8a^7/9$ and $Q = -16a^7/15 \neq 0$. The cube has genuine cubic anisotropy.

13 VOLUME INTEGRALS (CUBE)

13.1 Green's Tensor Integral Γ_0^{cube}

The zeroth-order integral $\int_{\text{cube}} G_{ij} \, d^3x = \Gamma_0^{\text{cube}} \, \delta_{ij}$ is still isotropic (only the second moment enters, and $\int x_i x_j \, d^3x \propto \delta_{ij}$ for the cube). The result differs from the sphere only through the geometry of the integration domain.

13.2 Second-Derivative Integral: Three Parameters

For the cube, the second-derivative integral acquires a third independent tensor structure:

$$\int_{\text{cube}} G_{ij,kl} \, d^3x = A^c \, \delta_{ij}\delta_{kl} + B^c \, (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) + C^c \, E_{ijkl}, \tag{35}$$

where C^c is the cubic anisotropy coefficient. Computing the three independent components:

$$A^c = I_{1122}, \qquad B^c = I_{1212}, \qquad C^c = I_{1111} - I_{1122} - 2\, I_{1212}. \tag{36}$$

SymPy computes $C^c = O\big((\omega a)^7\big)$, confirming it arises from the fourth-moment anisotropy of the cube.

14 T-MATRIX COUPLING COEFFICIENTS (CUBE)

The T-matrix tensor for the cube has cubic (not isotropic) symmetry:

$$T^c_{mnlp} = T^c_1 \, \delta_{mn}\delta_{lp} + T^c_2 \, (\delta_{ml}\delta_{np} + \delta_{mp}\delta_{nl}) + T^c_3 \, E_{mnlp}. \tag{37}$$

The third coefficient T^c_3 arises from C^c and vanishes for a sphere ($C = 0 \Rightarrow T_3 = 0$).

15 SELF-CONSISTENT SOLUTION: THE 6×6 VOIGT MATRIX AND ITS EIGENVALUE DECOMPOSITION

15.1 The Voigt Representation

The self-consistent strain equation $\varepsilon_{mn} = \varepsilon^0_{mn} + T_{mnlp}\,\varepsilon_{lp}$ is a 3×3 symmetric tensor equation. In Voigt notation, the six independent strain components

$$\boldsymbol{\varepsilon}_V = [\varepsilon_{11},\ \varepsilon_{22},\ \varepsilon_{33},\ 2\varepsilon_{23},\ 2\varepsilon_{13},\ 2\varepsilon_{12}]^T \tag{38}$$

are mapped to a 6-vector, and T_{mnlp} becomes a 6×6 matrix $\mathbf{T}_V$.

For the cubic T-tensor (37), $\mathbf{T}_V$ has the block-diagonal structure:

$$\mathbf{T}_V = \begin{pmatrix} \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{S} \end{pmatrix}, \tag{39}$$

where the 3×3 diagonal-strain block is

$$\mathbf{D} = T_1\,\mathbf{J} + (2T_2 + T_3)\,\mathbf{I}_3 = \begin{pmatrix} T_1 + 2T_2 + T_3 & T_1 & T_1 \\ T_1 & T_1 + 2T_2 + T_3 & T_1 \\ T_1 & T_1 & T_1 + 2T_2 + T_3 \end{pmatrix} \tag{40}$$

(with $\mathbf{J}$ the 3×3 matrix of ones) and the 3×3 off-diagonal shear block is

$$\mathbf{S} = 2T_2\,\mathbf{I}_3. \tag{41}$$

The physical reason for the block-diagonal structure is that the cubic anisotropy tensor E_{mnlp} only contributes when $m = n$ and $l = p$ (all indices equal). When either pair is off-diagonal ($m \neq n$ or $l \neq p$), $E_{mnlp} = 0$, so the T_3 term drops out and the off-diagonal shear components decouple with the isotropic eigenvalue $2T_2$.

15.2 Eigenvalue Decomposition: 4 Scalar Amplification Factors

The self-consistent equation $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^0 + \mathbf{T}_V\,\boldsymbol{\varepsilon}$ has the formal solution $\boldsymbol{\varepsilon} = (\mathbf{I}_6 - \mathbf{T}_V)^{-1}\,\boldsymbol{\varepsilon}^0$. Instead of inverting the full 6×6 matrix symbolically (which involves extremely expensive `simplify()` calls on nested rational expressions), we exploit the block structure to reduce everything to four scalar equations.

Block S (off-diagonal shear): Already diagonal. The amplification factor is

$$A_e^{(\text{off})} = \frac{1}{1 - 2T_2}. \tag{42}$$

Block D (diagonal strains): The matrix $\mathbf{D} = T_1\,\mathbf{J} + (2T_2 + T_3)\,\mathbf{I}_3$ has the well-known eigenstructure:

- **Dilatation mode:** eigenvector $[1, 1, 1]^T$ (uniform expansion), eigenvalue $3T_1 + 2T_2 + T_3$:

$$A_\theta = \frac{1}{1 - 3T_1 - 2T_2 - T_3}. \tag{43}$$

- **Diagonal deviatoric modes:** eigenvectors $[1, -1, 0]^T$ and $[1, 1, -2]^T$ (traceless diagonal strain), eigenvalue $2T_2 + T_3$:

$$A_e^{(\text{diag})} = \frac{1}{1 - 2T_2 - T_3}. \tag{44}$$

Plus the displacement amplification (unchanged from the sphere):

$$A_u = \frac{1}{1 - \omega^2 \Delta\rho\, \Gamma_0^{\text{cube}}}. \tag{45}$$

For the sphere ($T_3 = 0$): $A_e^{(\text{off})} = A_e^{(\text{diag})} \equiv A_e$ and we recover the 3 amplification factors of Section 8.

16 EFFECTIVE CONTRASTS (CUBE): SCALAR DERIVATION

The effective stiffness contrast is $\Delta c^*_{mnlp} = \Delta c_{mnuv}\, A_{uv,lp}$, where A is the amplification operator. We avoid the full 6×6 matrix product by decomposing the incident strain into the three eigenspaces of A and applying each scalar amplification factor separately.

The isotropic contrast $\Delta c_{mnuv}\, \varepsilon^*_{uv} = \Delta\lambda\, \theta^*\, \delta_{mn} + 2\Delta\mu\, \varepsilon^*_{mn}$. Substituting the amplified strain:

$$\varepsilon^*_{mn} = \underbrace{\frac{A_\theta\, \theta^0}{3}\, \delta_{mn}}_{\text{dilatation}} + \underbrace{A_e^{(\text{diag})}\, e^{0,\text{diag}}_{mn}}_{\text{diag. deviatoric}} + \underbrace{A_e^{(\text{off})}\, e^{0,\text{off}}_{mn}}_{\text{off-diag. shear}} \tag{46}$$

gives, after algebra:

Effective density:

$$\Delta\rho^{*,c} = \frac{\Delta\rho}{1 - \omega^2 \Delta\rho\, \Gamma_0^{\text{cube}}}. \tag{47}$$

Off-diagonal effective shear modulus:

$$\Delta\mu^{*,\text{off}} = \frac{\Delta\mu}{1 - 2T_2^c}. \tag{48}$$

Diagonal effective shear modulus:

$$\Delta\mu^{*,\text{diag}} = \frac{\Delta\mu}{1 - 2T_2^c - T_3^c}. \tag{49}$$

Effective Lamé parameter:

$$\Delta\lambda^{*,c} = \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{1 - 3T_1^c - 2T_2^c - T_3^c} - \frac{\Delta\mu}{3}\left(\frac{1}{1 - 2T_2^c - T_3^c} + \frac{1}{1 - 2T_2^c}\right). \tag{50}$$

The cubic anisotropy of the effective shear modulus is:

$$\Delta\mu^{*,\text{diag}} - \Delta\mu^{*,\text{off}} = \frac{\Delta\mu\, T_3^c}{(1-2T_2^c)(1-2T_2^c-T_3^c)} = O\big((\omega a)^7\big). \tag{51}$$

Part III: From Single Scatterer to Effective Medium

17 HOMOGENISATION AND EFFECTIVE MEDIUM THEORY

17.1 The Space-Filling Requirement

The ultimate goal of computing single-scatterer T-matrices is to build effective medium theories for heterogeneous elastic materials. The idea is simple: model the heterogeneous medium as a collection of scatterers, each characterised by its T-matrix, and then average over the ensemble to obtain effective material properties.

The critical requirement for such a homogenisation procedure is that the scatterers must *tile* the medium completely. If they do not, the material between scatterers is unmodelled, leading to systematic errors (Willis 1977; Hashin 1983).

Cubes are the simplest three-dimensional shape that tiles $\mathbb{R}^3$ without gaps or overlaps. This was famously proven to be optimal for convex shapes by the Keller–Minkowski theorem, and the practical importance for effective medium theory was recognised early on by Hashin and Shtrikman (Hashin & Shtrikman 1962a,b, 1963), who used cubic partitions as the foundation of their variational bounds on effective elastic moduli.

17.2 From T-Matrix to Effective Moduli

Given the cubic T-matrix (37), we can construct the effective elastic moduli of a composite material as follows. Consider a background medium partitioned into N non-overlapping cubes, each with (possibly different) material properties. In the long-wavelength limit, the effective stiffness tensor satisfies (Zeller & Dederichs 1973; Willis 1977):

$$C_{ijkl}^{\text{eff}} = C_{ijkl}^{(0)} + \frac{1}{V}\sum_{n=1}^{N} \Delta c_{ijuv}^{(n)}\, A_{uv,kl}^{(n)}, \tag{52}$$

where $C^{(0)}$ is the background stiffness, $\Delta c^{(n)}$ is the contrast of the n-th cube, and $A^{(n)}$ is the amplification operator derived from its T-matrix. In the dilute limit (non-interacting scatterers), each $A^{(n)}$ is computed independently as in Section 15.

17.3 Self-Consistent Effective Medium Schemes

More sophisticated approximations account for scatterer-scatterer interactions:

Coherent Potential Approximation (CPA): Each cube is embedded in the (unknown) effective medium rather than the background, leading to a self-consistent equation for C^{eff} (Zeller & Dederichs 1973; Gubernatis et al. 1977). The T-matrix formulation is particularly natural here: the CPA condition is simply that the average T-matrix vanishes, $\langle T \rangle = 0$.

Differential Effective Medium (DEM): Scatterers are added incrementally to the background, with the effective medium updated at each step (Norris 1985). The cubic T-matrix provides the "update kernel" for each infinitesimal addition.

Kramers–Kronig consistency: Because the T-matrix is derived from the full self-consistent Lippmann–Schwinger equation (not the Born approximation), the resulting effective moduli are causal and satisfy Kramers–Kronig relations (Willis 1981), making them suitable for broadband modelling.

17.4 Cubic Symmetry of the Effective Medium

Even when the individual cubes have isotropic material properties ($\Delta\lambda$, $\Delta\mu$, $\Delta\rho$), the effective medium inherits *cubic symmetry* from the scatterer geometry. This is a direct consequence of $T_3 \neq 0$ in (37): the effective shear modulus splits into $\mu^{*,\text{diag}}$ and $\mu^{*,\text{off}}$, giving the effective stiffness tensor three independent elastic constants (bulk modulus, and two shear moduli) rather than the two of an isotropic medium.

The cubic anisotropy parameter (51) quantifies this splitting. At low frequencies it is $O(\omega a)^7$—very small—but it grows rapidly with frequency, becoming significant as ka approaches unity.

17.5 Connection to Hashin–Shtrikman Bounds

The classical Hashin–Shtrikman variational bounds (Hashin & Shtrikman 1962a,b, 1963) on effective elastic moduli are derived using a comparison medium and a polarisation stress field. The bounds are achieved by specific microstructural geometries: the Hashin assemblage (concentric spheres) achieves the bulk modulus bound, but the shear modulus bound requires more complex geometries.

The cubic T-matrix approach provides a complementary, wave-based route to effective properties that naturally includes frequency dependence and finite-wavelength corrections absent from static variational bounds. In the zero-frequency limit, the cubic effective moduli computed here should lie within the Hashin–Shtrikman bounds, providing a consistency check.

18 BORN VERIFICATION (CUBE)

In the Born limit $a \to 0$:

$$\lim_{a\to 0} \Delta\rho^{*,c} = \Delta\rho, \qquad \lim_{a\to 0} \Delta\lambda^{*,c} = \Delta\lambda, \qquad \lim_{a\to 0} \Delta\mu^{*,\text{diag}} = \Delta\mu, \qquad \lim_{a\to 0} \Delta\mu^{*,\text{off}} = \Delta\mu. \tag{53}$$

The cubic anisotropy vanishes: $\lim_{a\to 0}(\Delta\mu^{*,\text{diag}} - \Delta\mu^{*,\text{off}}) = 0$. All limits verified analytically with SymPy.

Part IV: The Global T-Matrix

19 FROM LOCAL SITE T-MATRIX TO GLOBAL MULTIPLE SCATTERING

The T-matrix $\mathbf{T}_0$ derived in Parts I and II is a *local site* (or single-scatterer) quantity: it describes the self-consistent response of one cubic cell in isolation, embedded in a homogeneous reference medium. To model a heterogeneous medium composed of many such cells, we must account for the multiple scattering between sites—each cube scatters incident waves, which propagate to neighbouring cubes, scatter again, and so on.

The standard framework for this is the Foldy–Lax multiple-scattering theory (Foldy 1945; Lax 1952), recast in the T-matrix formalism by Waterman (1976) and applied to elastic media by Zeller & Zeller & Dederichs (1973), Gubernatis et al. (1977), and Jakobsen (2012).

19.1 The Lippmann–Schwinger Equation on a Lattice

Partition the medium into N non-overlapping cubes centred at positions $\{\mathbf{x}_n\}_{n=1}^N$, each with its own material contrasts $\Delta c^{(n)}$, $\Delta\rho^{(n)}$. The total scattered field at site m due to all other sites is:

$$u_i^{\text{tot}}(\mathbf{x}_m) = u_i^0(\mathbf{x}_m) + \sum_{n\neq m} G_{ij}^{(0)}(\mathbf{x}_m - \mathbf{x}_n)\, \mathbf{T}_0^{(n)} \cdot u_j^{\text{tot}}(\mathbf{x}_n), \tag{54}$$

where $G_{ij}^{(0)}(\mathbf{x}_m - \mathbf{x}_n)$ is the background (reference medium) Green's tensor evaluated at the vector separation between the centres of cubes m and n, and $\mathbf{T}_0^{(n)}$ is the local site T-matrix for the n-th cube (computed as in Part II).

This is the elastic analogue of the Foldy–Lax equations (Foldy 1945; Lax 1952). In compact operator notation, writing $\mathbf{u} = [u(\mathbf{x}_1), \ldots, u(\mathbf{x}_N)]^T$ and assembling the inter-site Green's tensors into a matrix $\mathbf{G}_0$:

$$\mathbf{u} = \mathbf{u}^0 + \mathbf{G}_0\, \mathbf{T}_0\, \mathbf{u}, \tag{55}$$

where $\mathbf{T}_0 = \text{diag}(\mathbf{T}_0^{(1)}, \ldots, \mathbf{T}_0^{(N)})$ is the block-diagonal matrix of local site T-matrices.

19.2 The Global T-Matrix

Solving (55) for $\mathbf{u}$ gives $\mathbf{u} = (\mathbf{I} - \mathbf{G}_0\,\mathbf{T}_0)^{-1}\,\mathbf{u}^0$. The global T-matrix is defined as the operator that maps the incident field to the total scattered field:

$$\mathbf{T} = \mathbf{T}_0\,(\mathbf{I} - \mathbf{G}_0\,\mathbf{T}_0)^{-1}. \tag{56}$$

Equivalently, by pre-multiplying with $\mathbf{G}_0$:

$$\mathbf{G}_0\,\mathbf{T} = \mathbf{G}_0\,\mathbf{T}_0\,(\mathbf{I} - \mathbf{G}_0\,\mathbf{T}_0)^{-1} = (\mathbf{I} - \mathbf{G}_0\,\mathbf{T}_0)^{-1} - \mathbf{I}. \tag{57}$$

19.3 Physical Interpretation: The Neumann Series

When $\|\mathbf{G}_0\,\mathbf{T}_0\| < 1$ (weak scattering), the inverse can be expanded as a Neumann series:

$$\mathbf{T} = \mathbf{T}_0 + \mathbf{T}_0\,\mathbf{G}_0\,\mathbf{T}_0 + \mathbf{T}_0\,\mathbf{G}_0\,\mathbf{T}_0\,\mathbf{G}_0\,\mathbf{T}_0 + \cdots \tag{58}$$

Each term has a transparent physical interpretation (Figure 1):

- **Zeroth order** $\mathbf{T}_0$: Single scattering at each site (the Born–T-matrix approximation).
- **First order** $\mathbf{T}_0\,\mathbf{G}_0\,\mathbf{T}_0$: A wave scatters at site n ($\mathbf{T}_0^{(n)}$), propagates to site m ($\mathbf{G}_0$), and scatters again ($\mathbf{T}_0^{(m)}$).
- k**-th order:** Chains of $(k+1)$ scattering events connected by k propagation steps.

The full inverse $(\mathbf{I} - \mathbf{G}_0\,\mathbf{T}_0)^{-1}$ sums this series to all orders, capturing the complete multiple-scattering response.

[Schematic to be added.] A pictorial representation showing: (a) a regular lattice of cubes, each labelled with its local T-matrix $\mathbf{T}_0^{(n)}$; (b) arrows representing the inter-site Green's tensor $G_{ij}^{(0)}(\mathbf{x}_m - \mathbf{x}_n)$ connecting cube centres; (c) the multiple-scattering chain: incident wave $\to$ scatter at site $n \to$ propagate $\to$ scatter at site $m \to$ propagate $\to \cdots$

Figure 1: Schematic of the global T-matrix formalism. Left: the medium is partitioned into cubes, each with a local T-matrix $\mathbf{T}_0^{(n)}$. Right: the inter-site Green's tensor $\mathbf{G}_0$ propagates the scattered field between cube centres, and the global T-matrix (56) sums all multiple-scattering paths to all orders.

20 THE INTER-SITE GREEN'S TENSOR $\mathbf{G}_0$

20.1 Homogeneous Reference Medium

In a homogeneous isotropic reference medium, the inter-site Green's tensor is the free-space Green's tensor evaluated at the separation vector:

$$G_{ij}^{(0)}(\mathbf{x}_m - \mathbf{x}_n) = f(|\mathbf{x}_m - \mathbf{x}_n|)\,\delta_{ij} + g(|\mathbf{x}_m - \mathbf{x}_n|)\,\hat{r}_{mn,i}\,\hat{r}_{mn,j}, \tag{59}$$

where $f(r)$ and $g(r)$ are the radial functions from Section 3, and $\hat{r}_{mn,i} = (x_{m,i} - x_{n,i})/|\mathbf{x}_m - \mathbf{x}_n|$ is the unit vector from cube n to cube m.

For a regular cubic lattice with spacing $2a$ (so that adjacent cube faces are touching), the inter-site vectors are $\mathbf{x}_m - \mathbf{x}_n = 2a\,(p, q, r)$ for integers (p, q, r) with $(p, q, r) \neq (0, 0, 0)$.

20.2 Layered Reference Medium: The Kennett Propagator

For geophysical applications, the background medium is not homogeneous but *stratified*—consisting of horizontal layers with depth-dependent properties. In this case, the inter-site Green's tensor must be replaced by the Green's tensor for the layered reference medium.

Kennett (1983) showed that the elastic Green's tensor for a plane-layered medium can be computed efficiently using the reflectivity method: the wavefield is decomposed into plane waves (via horizontal wavenumber integration), each of which propagates through the layer stack via a system of reflection and transmission matrices. The key recursive relations are:

(i) **Layer propagator:** For a homogeneous layer of thickness h with P- and S-wave vertical slownesses q_α and q_β, the phase propagation across the layer is given by $\mathbf{E}(h) = \mathrm{diag}(e^{i\omega q_\alpha h}, e^{i\omega q_\beta h})$.

(ii) **Interface coefficients:** At each interface, Zoeppritz-type reflection and transmission coefficients r_d, r_u, t_d, t_u (for both P–P, P–SV, SV–P, SV–SV conversions) relate the up- and down-going wave amplitudes.

(iii) **Kennett recursive algorithm:** The generalised reflection matrix R_D for a stack of layers is built recursively from the bottom up:

$$R_D^{(j)} = r_d^{(j)} + t_u^{(j)}\, R_D^{(j+1)} \left(\mathbf{I} - r_u^{(j)}\, R_D^{(j+1)}\right)^{-1} t_d^{(j)}, \tag{60}$$

where $r_d^{(j)}, t_d^{(j)}$ are the reflection/transmission matrices at the j-th interface, and $R_D^{(j+1)}$ is the already-computed reflection matrix for all layers below interface j (Kennett 1983, 1979).

(iv) **Green's tensor assembly:** The frequency–wavenumber-domain Green's tensor is assembled from the reflection matrices and layer propagators. A final inverse Hankel (or Fourier) transform over horizontal wavenumber yields $G_{ij}^{(0)}(\mathbf{x}_m - \mathbf{x}_n; \omega)$ in the space-frequency domain.

20.3 Significance for Heterogeneous Media Modelling

The combination of the Kennett propagator for $\mathbf{G}_0$ with the cubic local T-matrix $\mathbf{T}_0$ provides a powerful framework for modelling wave propagation in realistic heterogeneous Earth models:

(i) **Background model:** A plane-layered velocity-density profile (e.g. from well logs or a 1D Earth

model such as PREM or AK135) defines the reference medium. The Green's tensor $\mathbf{G}_0$ for this layered background is computed using the Kennett reflectivity method.

(ii) **3D heterogeneity:** The departures from the layered background are partitioned into cubic cells, each with constant (but arbitrary) material contrasts $\Delta\lambda^{(n)}$, $\Delta\mu^{(n)}$, $\Delta\rho^{(n)}$. The local T-matrix $\mathbf{T}_0^{(n)}$ for each cube is computed analytically (Part II)—it depends only on the cube's material contrasts and the local background properties.

(iii) **Multiple scattering:** The global T-matrix (56) is assembled from the $\mathbf{T}_0^{(n)}$ and $\mathbf{G}_0$, and the total scattered field is obtained by applying $\mathbf{T}$ to the incident field.

This is essentially the approach of Jakobsen (2012) and Jakobsen & Wu (2016), but with the important refinement that the local T-matrix is computed self-consistently (not in the Born approximation), allowing strong contrasts to be modelled accurately.

21 OPERATOR STRUCTURE AND SYMMETRIES

21.1 Block Structure of the Global System

The global system (55) has a natural block structure. For N cubes, each described by 9 degrees of freedom (3 displacement + 6 strain components), the system matrices are of size $9N \times 9N$. However, the local T-matrix $\mathbf{T}_0$ is block-diagonal (each block is 9×9), and the inter-site Green's tensor $\mathbf{G}_0$ has translational structure (for a regular lattice, G_{mn} depends only on $\mathbf{x}_m - \mathbf{x}_n$).

For a regular cubic lattice, this translational invariance can be exploited: the system (55) becomes block-circulant and can be diagonalised by a spatial Fourier transform over the lattice. In Fourier space, the global T-matrix at wavevector $\mathbf{k}$ reduces to:

$$\hat{\mathbf{T}}(\mathbf{k}) = \hat{\mathbf{T}}_0 \left(\mathbf{I} - \hat{\mathbf{G}}_0(\mathbf{k})\, \hat{\mathbf{T}}_0\right)^{-1}, \tag{61}$$

where $\hat{\mathbf{G}}_0(\mathbf{k}) = \sum_{\mathbf{x}_n \neq \mathbf{0}} \mathbf{G}_0(\mathbf{x}_n)\, e^{-i\mathbf{k}\cdot\mathbf{x}_n}$ is the lattice Green's function (also called the structure factor of the Green's tensor).

This Fourier decomposition reduces the $9N \times 9N$ system to N independent 9×9 systems, one for each wavevector in the first Brillouin zone of the cubic lattice—a dramatic computational saving.

21.2 Relationship to the Dyson Equation

Equation (56) is the lattice analogue of the Dyson equation from many-body theory. Defining the self-energy (or mass operator) as $\Sigma = \mathbf{T}_0$, the effective Green's function of the heterogeneous medium satisfies:

$$\mathbf{G}_{\text{eff}} = \mathbf{G}_0 + \mathbf{G}_0\, \Sigma\, \mathbf{G}_{\text{eff}} = (\mathbf{I} - \mathbf{G}_0\, \Sigma)^{-1}\, \mathbf{G}_0. \tag{62}$$

The poles of $\mathbf{G}_{\text{eff}}$ in the frequency–wavenumber domain give the dispersion relations of the effective medium, including attenuation due to scattering. This connection to the Dyson equation was exploited by Zeller & Zeller & Dederichs (1973) for elastic media and by Jakobsen (2012) for seismic modelling.

22 COMPUTATIONAL METHODS FOR INVERTING THE GLOBAL SYSTEM

The central computational problem is to solve the linear system

$$(\mathbf{I} - \mathbf{G}_0\,\mathbf{T}_0)\,\mathbf{u} = \mathbf{u}^0, \tag{63}$$

which is of dimension $9N \times 9N$ (three displacement and six independent strain components per cube, N cubes). For realistic 3D models, N may range from 10^3 (small exploration-scale problems) to 10^6 or beyond (crustal-scale), making the choice of solution method critical.

We survey five approaches, ranging from the simplest (direct inversion, the Neumann series) to the most sophisticated (fast multipole methods). Each has a distinct domain of applicability.

22.1 Direct Inversion

The most straightforward approach is to form and factorise $(\mathbf{I}-\mathbf{G}_0\,\mathbf{T}_0)$ explicitly, using LU decomposition or a similar dense solver. This costs $O(N^3)$ in time and $O(N^2)$ in storage, restricting it to small problems ($N \lesssim 10^3$). However, it yields the complete resolvent, from which the Green's function for all source–receiver pairs can be extracted. For benchmark problems and validation of approximate methods, direct inversion remains indispensable.

22.2 The Neumann (Born) Series

The Neumann series (58) is a fixed-point iteration:

$$\mathbf{u}^{(k+1)} = \mathbf{u}^0 + \mathbf{G}_0\,\mathbf{T}_0\,\mathbf{u}^{(k)}, \qquad \mathbf{u}^{(0)} = \mathbf{u}^0. \tag{64}$$

It converges if and only if the spectral radius $\rho(\mathbf{G}_0\,\mathbf{T}_0) < 1$. In scattering theory this is the Born series, and it is well known to diverge for strong material contrasts (Gubernatis et al. 1977). Even when it converges, the rate is geometric with ratio ρ, so many iterations may be needed for moderate contrasts. The Neumann series is therefore useful mainly as a conceptual tool and as the starting point for more sophisticated iterative methods.

22.3 Iterative Krylov Subspace Methods

Krylov subspace methods—GMRES (Saad & Schultz 1986), BiCGSTAB (van der Vorst 1992), and their variants—are the practical workhorse for solving (63). They require only the matrix–vector product $\mathbf{v} \mapsto \mathbf{G}_0\,\mathbf{T}_0\,\mathbf{v}$, never the explicit matrix, and achieve convergence in far fewer iterations than the Neumann series because they minimise the residual over the full Krylov subspace rather than iterating a single fixed-point map.

GMRES (Generalised Minimal Residual) (Saad & Schultz 1986) is the method of choice for non-symmetric systems. At iteration k it minimises $\|\mathbf{r}^{(k)}\|$ over the Krylov subspace $\mathcal{K}_k = \mathrm{span}\{\mathbf{r}^{(0)}, \mathbf{A}\mathbf{r}^{(0)}, \ldots, \mathbf{A}^{k-1}\mathbf{r}^{(0)}\}$, where $\mathbf{A} = \mathbf{I} - \mathbf{G}_0\,\mathbf{T}_0$. Storage grows linearly with k (one vector per iteration), which can be controlled by restarting (GMRES(m)).

BiCGSTAB (van der Vorst 1992) is an alternative that avoids the growing storage of GMRES at the cost of a less smooth convergence history. Each iteration requires two matrix–vector products but only a fixed amount of storage.

The matrix–vector product $\mathbf{G}_0\,\mathbf{T}_0\,\mathbf{v}$ has a transparent physical interpretation: (i) *scatter*: apply the block-diagonal $\mathbf{T}_0$ to $\mathbf{v}$ (cost $O(N)$, since each 9×9 block acts independently); (ii) *propagate*: apply the dense inter-site Green's tensor $\mathbf{G}_0$ (cost $O(N^2)$ naively, but reducible to $O(N \log N)$ with the methods of Section 22.5).

Preconditioning. The convergence rate of Krylov methods depends on the spectral properties of $\mathbf{I} - \mathbf{G}_0\,\mathbf{T}_0$. Because the local T-matrix $\mathbf{T}_0$ already absorbs the strong single-site scattering self-consistently, the remaining inter-site coupling $\mathbf{G}_0\,\mathbf{T}_0$ is typically much weaker than the original material contrast, leading to favourable convergence even without explicit preconditioning. When additional acceleration is needed, block-diagonal or block-Jacobi preconditioners (using the diagonal blocks of $\mathbf{G}_0$) are natural choices.

Historical note. Krylov methods were applied to elastic-wave scattering in layered media with superimposed spherical heterogeneities by Nestor (1996), who solved the multiple-scattering series for the global T-matrix using iterative methods with the Kennett reflectivity method (Section 20.2) providing the inter-site Green's tensor $\mathbf{G}_0$ and spherical scatterer T-matrices for $\mathbf{T}_0$. This combination demonstrated the feasibility of modelling 3D heterogeneous media as perturbations to a stratified background, but was limited by the isotropy of the spherical T-matrix and the inability of spheres to tile space. The cubic T-matrix developed in Part II removes both limitations, completing the bridge from single-scatterer physics to rigorous homogenisation that was implicit in the original formulation.

22.4 The Renormalised Scattering Series

Jakobsen & Wu (2016) proposed a renormalised scattering series that overcomes the convergence

limitations of the Born/Neumann series for strong contrasts. The idea is to split the total perturbation into a "strong" part (absorbed into a modified reference medium or into the local T-matrix) and a "weak" residual, then iterate on the residual only.

Concretely, define a modified operator $\tilde{\mathbf{G}}_0 = \mathbf{G}_0 - \mathbf{G}_0^{\text{diag}}$, where $\mathbf{G}_0^{\text{diag}}$ contains only the self-interaction terms (which are already accounted for in $\mathbf{T}_0$). The renormalised series iterates:

$$\mathbf{u}^{(k+1)} = \mathbf{u}^0 + \tilde{\mathbf{G}}_0\, \mathbf{T}_0\, \mathbf{u}^{(k)}. \tag{65}$$

Because $\tilde{\mathbf{G}}_0$ excludes the (already-resummed) on-site terms, the spectral radius $\rho(\tilde{\mathbf{G}}_0\, \mathbf{T}_0)$ is typically much smaller than $\rho(\mathbf{G}_0\, \mathbf{T}_0)$, extending the domain of convergence to stronger contrasts. When combined with the self-consistent (non-Born) local T-matrix, this approach effectively sums the dominant scattering contributions exactly and treats only the residual inter-site coupling perturbatively.

The renormalised series can also be viewed as a preconditioned Neumann iteration, and it can be further accelerated by embedding it within a Krylov solver.

22.5 Fast Multipole and Hierarchical Matrix Methods

The dominant cost of each Krylov iteration is the dense matrix–vector product $\mathbf{G}_0\, \mathbf{v}$, which costs $O(N^2)$ when computed directly. For large N, this can be reduced to $O(N \log N)$ or even $O(N)$ by exploiting the spatial structure of the Green's tensor.

Fast Multipole Method (FMM): The elastic Green's tensor $G_{ij}^{(0)}(\mathbf{x})$ decays as $1/|\mathbf{x}|$ and is smooth (low-rank) when source and receiver are well separated. The FMM (Rokhlin 1985; Greengard & Rokhlin 1987) exploits this by grouping cubes into a hierarchical tree structure, computing far-field interactions via multipole expansions, and only evaluating near-field interactions directly. For elastodynamics, the FMM was developed by Fujiwara (2000) and Chaillat et al. (2008), achieving $O(N \log N)$ or $O(N)$ complexity per matrix–vector product.

Hierarchical matrices ($\mathcal{H}$-matrices): An alternative to the FMM, $\mathcal{H}$-matrices (Hackbusch 1999; Börm et al. 2003) represent the off-diagonal blocks of $\mathbf{G}_0$ in low-rank factored form, again achieving $O(N \log N)$ storage and matrix–vector products. The advantage over FMM is that $\mathcal{H}$-matrices can also be inverted or LU-factored in $O(N \log^2 N)$ time, providing a direct (non-iterative) solver.

Hybrid FFT for layered $\mathbf{G}_0$: When the reference medium is layered (Section 20.2) and the cubes are arranged on a regular horizontal grid (but at variable depths), the horizontal convolution structure of $\mathbf{G}_0$ can be exploited via 2D FFT. Only the vertical coupling between different depth levels requires explicit summation. This hybrid approach naturally suits the "layered background + 3D perturbations" geometry of geophysical applications.

[ht]

Method	Cost	Requires	Best for
Direct inversion	$O(N^3)$	Dense matrix	Small N, benchmarks
Neumann series	$O(kN^2)$	$\rho < 1$	Weak scattering only
GMRES/BiCGSTAB	$O(kN^2)$ or $O(kN\log N)$	Mat-vec product	General heterogeneity
Renormalised series	$O(kN^2)$ or $O(kN\log N)$	Self-consist. T_0	Strong contrasts
FMM / $\mathcal{H}$-matrices	$O(N\log N)$ per iter.	Spatial decay of G_0	Large N, 3D

Table 1. Comparison of computational methods for solving the global T-matrix system (63). Here k denotes the number of iterations required for convergence.

22.6 Summary and Recommendations

Table 1 summarises the computational methods and their domains of applicability.

For the geophysical application envisioned here—3D heterogeneities superimposed on a layered background via the Kennett propagator—the recommended approach is a Krylov solver (GMRES or BiCGSTAB) with the matrix–vector product $\mathbf{G}_0\,\mathbf{T}_0\,\mathbf{v}$ accelerated by the hybrid FFT/direct scheme described in Section 22.5. The self-consistent local T-matrix $\mathbf{T}_0$ from Part II serves as a natural preconditioner by absorbing the strong single-site scattering, leaving only the weaker inter-site coupling to be resolved iteratively.

Generated from `tmatrix_analytic.py` *and* `tmatrix_cube.py` *using SymPy symbolic computation. All sphere integrals evaluated in exact closed form; cube integrals evaluated semi-analytically via Taylor expansion to* $O(\omega a)^8$.

Part V: Numerical Implementation

23 HYBRID COMPUTATION STRATEGY

The volume integrals derived in Parts I and II can be evaluated analytically by SymPy (symbolic computer algebra) for both the sphere and cube geometries. However, for deployment in a numerical wavefield-modelling code, a pure-NumPy implementation is desirable. This section describes the hybrid numerical strategy used in the `CubicTMatrix` Python package.

23.1 The Singularity Structure of the Integrands

The two volume integrals that define the local site T-matrix have fundamentally different singularity structures at the origin:

Green's tensor integral (Eq 18 / Section 13.1): The integrand $G_{ij}(\mathbf{x}) \sim 1/r$ as $r \to 0$. In

three dimensions, $d^3x \sim r^2\,dr$, so the integral converges absolutely. Standard 3D Gauss–Legendre quadrature evaluates it accurately.

Second-derivative integral (Eq 35): The integrand $G_{ij,kl}(\mathbf{x}) \sim 1/r^3$ as $r \to 0$. This is the static Eshelby singularity. The volume integral *diverges*—it must be understood as a Cauchy principal value plus a shape-dependent depolarisation tensor. Direct numerical quadrature captures the divergent static part with an n-dependent finite remainder, producing incorrect results.

23.2 The Hybrid Approach

The solution is to use a different method for each integral:

- Γ_0**: full numerical quadrature.** Vectorised 3D Gauss–Legendre quadrature over $[-a, a]^3$ with $n_{\mathrm{GL}} = 32$ points per dimension. The GL nodes never include the origin (they are the zeros of Legendre polynomials on $(-1, 1)$), and the $1/r$ singularity is integrable, so the integral converges rapidly. This correctly captures both the real (static) and imaginary (frequency-dependent) contributions.
- A^c**,** B^c**,** C^c**: polynomial Taylor expansion.** We decompose the Green's tensor into a singular static part ($1/r^3$) and a smooth polynomial remainder. Only the smooth remainder is integrated—the static singularity drops out automatically because it is excluded from the Taylor series of the frequency-dependent terms. The polynomial integrand is evaluated by GL quadrature, which integrates it exactly (to Taylor truncation order).

This hybrid strategy achieves the accuracy of the SymPy symbolic computation (matching T_1 and T_2 to $\sim 10^{-8}$, T_3 to $\sim 0.3\%$) while running in pure NumPy at negligible computational cost.

24 SMOOTH GREEN'S TENSOR DECOMPOSITION

24.1 Splitting off the Singularity

The Green's tensor (Eq 4) can be decomposed as:

$$G_{ij}(\mathbf{x}) = G_{ij}^{\mathrm{static}}(\mathbf{x}) + G_{ij}^{\mathrm{s}}(\mathbf{x}), \tag{66}$$

where the static part $G_{ij}^{\mathrm{static}} \sim 1/r$ produces the divergent $1/r^3$ second derivatives, and the smooth part G_{ij}^{s} is an entire function of $\mathbf{x}$ (a convergent power series).

The smooth part arises from the odd-power terms in the Taylor expansion of $f(r) \cdot r$ and $g(r) \cdot r$ about $r = 0$. It has the structure:

$$\boxed{G_{ij}^{\mathrm{s}}(\mathbf{x}) = \delta_{ij}\,\Phi(r^2) + x_i\,x_j\,\Psi(r^2)} \tag{67}$$

where $\Phi(u) = \sum_n \phi_n\,u^n$ and $\Psi(u) = \sum_n \psi_n\,u^n$ are power series in $u = r^2$.

24.2 Taylor Coefficients

The Taylor coefficients are derived by expanding the exponentials $e^{i\omega r/\alpha}$ and $e^{i\omega r/\beta}$ in the definitions of $f(r)$ and $g(r)$ (Eqs 5–6), multiplying by r, and collecting terms of odd power in r (which become polynomial after division by r):

$$\boxed{\phi_n = \frac{(i\omega/\beta)^{2n+1}}{(2n+1)!\,4\pi\rho\beta^2}} \tag{68}$$

$$\boxed{\psi_n = \frac{1}{4\pi\rho}\left[\frac{(i\omega/\alpha)^{2n+3}}{\alpha^2} - \frac{(i\omega/\beta)^{2n+3}}{\beta^2}\right]\frac{1}{(2n+3)!}} \tag{69}$$

In practice, $N_{\text{Taylor}} = 8$ terms suffice for convergence at the frequencies of interest ($\omega a/\beta < 1$).

24.3 Smooth Second Derivatives

The second spatial derivatives of G^{s}_{ij} are computed via the chain rule, using $\partial r^2/\partial x_k = 2x_k$. The three independent components needed for the cube's A, B, C decomposition (Eq 36) are:

$$G^{\text{s}}_{11,11} = 4x_1^2\,\Phi'' + 2\Phi' + 2\Psi + 10x_1^2\,\Psi' + 4x_1^4\,\Psi'', \tag{70}$$

$$G^{\text{s}}_{11,22} = 4x_2^2\,\Phi'' + 2\Phi' + 2x_1^2\,\Psi' + 4x_1^2x_2^2\,\Psi'', \tag{71}$$

$$G^{\text{s}}_{12,12} = \Psi + 2(x_1^2 + x_2^2)\,\Psi' + 4x_1^2x_2^2\,\Psi'', \tag{72}$$

where primes denote d/du with $u = r^2$. These are polynomial in (x_1, x_2, x_3)—completely free of singularities at the origin—and are integrated over the cube by GL quadrature.

The integrals give A^c, B^c, C^c via (Eq 36):

$$A^c = \int G^{\text{s}}_{11,22}\,d^3x, \quad B^c = \int G^{\text{s}}_{12,12}\,d^3x, \quad C^c = \int G^{\text{s}}_{11,11}\,d^3x - A^c - 2B^c. \tag{73}$$

25 DISPLACEMENT-TRACTION BASIS

For integration into the Kennett reflectivity propagator framework (Section 20.2), the T-matrix must operate on the displacement-traction state vector rather than the strain tensor. This section derives the 6×6 T-matrix in the $(u_z, u_x, u_y, t_{zz}, t_{xz}, t_{yz})$ basis.

25.1 Coordinate System

We adopt a right-handed coordinate system with z (down), x (right), y (out of page). The mapping from the generic $(1, 2, 3)$ Cartesian indices used in Parts I–IV to (z, x, y) is: $1 \to z$, $2 \to x$, $3 \to y$. The Voigt ordering is:

$$\boldsymbol{\varepsilon}_V = [\varepsilon_{zz},\ \varepsilon_{xx},\ \varepsilon_{yy},\ 2\varepsilon_{xy},\ 2\varepsilon_{zy},\ 2\varepsilon_{zx}]^T. \tag{74}$$

25.2 Strain Extraction from Displacement-Traction

For a plane wave with horizontal wavenumbers (k_x, k_y), the six independent strains can be expressed in terms of the six displacement-traction components *without requiring* k_z. The traction components absorb the vertical-wavenumber dependence through the stress-strain relation. The key relations are:

$$\varepsilon_{zz} = \frac{t_{zz} - \lambda\, ik_x\, u_x - \lambda\, ik_y\, u_y}{\lambda + 2\mu}, \tag{75}$$

$$\varepsilon_{xx} = ik_x\, u_x, \qquad \varepsilon_{yy} = ik_y\, u_y, \tag{76}$$

$$2\varepsilon_{zx} = \frac{t_{xz}}{\mu}, \qquad 2\varepsilon_{zy} = \frac{t_{yz}}{\mu}, \tag{77}$$

$$2\varepsilon_{xy} = ik_y\, u_x + ik_x\, u_y. \tag{78}$$

These define a 6×6 strain extraction matrix $\mathbf{S}$ such that $\boldsymbol{\varepsilon}_V = \mathbf{S}(k_x, k_y)\ \mathbf{v}^0$ where $\mathbf{v} = (u_z, u_x, u_y, t_{zz}, t_{xz}, t_{yz})^T$:

$$\mathbf{S} = \begin{pmatrix} 0 & -\dfrac{\lambda\, ik_x}{M} & -\dfrac{\lambda\, ik_y}{M} & \dfrac{1}{M} & 0 & 0 \\ 0 & ik_x & 0 & 0 & 0 & 0 \\ 0 & 0 & ik_y & 0 & 0 & 0 \\ 0 & ik_y & ik_x & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \dfrac{1}{\mu} \\ 0 & 0 & 0 & 0 & \dfrac{1}{\mu} & 0 \end{pmatrix}, \tag{79}$$

where $M = \lambda + 2\mu$. Note that u_z does not appear explicitly: the vertical strain ε_{zz} is extracted entirely from the traction t_{zz} and the horizontal displacement components.

25.3 Voigt T-Matrix

The 6×6 Voigt T-matrix (Eqs 39–41) in the (z, x, y) coordinate system is:

$$\mathbf{T}_V = \begin{pmatrix} \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_{\text{shear}} \end{pmatrix}, \qquad \mathbf{D} = T_1\, \mathbf{J} + (2T_2 + T_3)\, \mathbf{I}_3, \qquad \mathbf{S}_{\text{shear}} = 2T_2\, \mathbf{I}_3. \tag{80}$$

This maps Voigt strain $\rightarrow$ Voigt effective stress perturbation: $\Delta\boldsymbol{\sigma}_V^* = \mathbf{T}_V\, \boldsymbol{\varepsilon}_V^0$.

25.4 Effective Stiffness Contrast Matrix

The effective stiffness contrast tensor Δc^*_{mnlp} in Voigt form has cubic symmetry:

$$\Delta\mathbf{c}_V^* = \begin{pmatrix} \mathbf{D}^* & \mathbf{0} \\ \mathbf{0} & \mathbf{S}^* \end{pmatrix}, \tag{81}$$

where

$$D^*_{ii} = \Delta\lambda^* + 2\Delta\mu^{*,\mathrm{diag}}, \quad D^*_{ij} = \Delta\lambda^* \ (i \neq j), \quad \mathbf{S}^* = 2\Delta\mu^{*,\mathrm{off}}\,\mathbf{I}_3, \tag{82}$$

with the effective contrasts $\Delta\lambda^*$, $\Delta\mu^{*,\mathrm{diag}}$, $\Delta\mu^{*,\mathrm{off}}$ given by Eqs (48)–(50).

25.5 Traction Projection

The traction on a $z =$ const surface is extracted from Voigt stress by a 3×6 projection matrix $\mathbf{P}$:

$$\begin{pmatrix} t_{zz} \\ t_{xz} \\ t_{yz} \end{pmatrix} = \mathbf{P}\,\boldsymbol{\sigma}_V, \qquad \mathbf{P} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}, \tag{83}$$

where σ_{zx} is Voigt index 6 and σ_{zy} is Voigt index 5.

25.6 The 6×6 T-Matrix in the $(\mathbf{u}, \mathbf{t})$ Basis

The full T-matrix in the displacement-traction basis maps $(\mathbf{u}, \mathbf{t})^0 \to$ scattered $(\mathbf{u}, \mathbf{t})$. Its structure is:

$$\boxed{\mathbf{T}_{6\times6} = V\,\big[\text{density block} + \text{stiffness block}\big]} \tag{84}$$

The **density scattering** contribution to the scattered displacement is:

$$u_i^{\mathrm{scat}} = V\,\omega^2\,\Delta\rho^*\,\Gamma_0\,u_i^0, \tag{85}$$

giving a diagonal 3×3 block in the upper-left of $\mathbf{T}_{6\times6}$.

The **stiffness scattering** contribution to the scattered traction is:

$$\mathbf{t}^{\mathrm{scat}} = V\,\mathbf{P}\,\Delta\mathbf{c}^*_V\,\mathbf{S}(k_x, k_y)\,\mathbf{v}^0. \tag{86}$$

This fills the lower 3×6 block. The complete matrix is:

$$\mathbf{T}_{6\times6} = V \begin{pmatrix} \omega^2\,\Delta\rho^*\,\Gamma_0\,\mathbf{I}_3 & \mathbf{0}_3 \\ \mathbf{P}\,\Delta\mathbf{c}^*_V\,\mathbf{S}(k_x, k_y) & \end{pmatrix}. \tag{87}$$

The displacement scattering from stiffness contrasts (upper-right blocks) requires the strain-Green's tensor H_{ijk} evaluated at the observation point, which is handled by the propagation framework (FFT-Prop). At the local site, the self-consistent amplification factors (Eqs 42–45) already account for this coupling.

26 NUMERICAL EXAMPLE (CUBE — NUMPY)

Using the same parameters as Section 10: $V_P = 5000$ m/s, $V_S = 3000$ m/s, $\rho = 2500$ kg/m^3. Contrasts: $\Delta\lambda = 2$ GPa, $\Delta\mu = 1$ GPa, $\Delta\rho = 100$ kg/m^3. Frequency $f = 10$ Hz, cube half-width $a = 10$ m.

Quantity	**Value (NumPy hybrid method)**
$k_P a$	0.1257
$k_S a$	0.2094
Γ_0	$2.604567\text{e}{-09} + 4.347360\text{e}{-10}\,i$
A^c	$0.000000\text{e}{+00} - 8.607635\text{e}{-14}\,i$
B^c	$0.000000\text{e}{+00} + 3.973958\text{e}{-14}\,i$
C^c	$0.000000\text{e}{+00} - 1.044478\text{e}{-19}\,i$
T_1^c	$0.000000\text{e}{+00} + 2.252429\text{e}{-04}\,i$
T_2^c	$0.000000\text{e}{+00} - 4.633677\text{e}{-05}\,i$
T_3^c	$0.000000\text{e}{+00} - 2.088956\text{e}{-10}\,i$
A_u	$1.001029\text{e}{+00} + 1.719804\text{e}{-04}\,i$
A_θ	$9.999997\text{e}{-01} + 5.830547\text{e}{-04}\,i$
A_e^{off}	$1.000000\text{e}{+00} - 9.267354\text{e}{-05}\,i$
A_e^{diag}	$1.000000\text{e}{+00} - 9.267375\text{e}{-05}\,i$
$\Delta\rho^{*,c}$	$1.001029\text{e}{+02} + 1.719804\text{e}{-02}\,i$
$\Delta\lambda^{*,c}$	$1.999999\text{e}{+09} + 1.616595\text{e}{+06}\,i$
$\Delta\mu^{*,\text{diag}}$	$1.000000\text{e}{+09} - 9.267375\text{e}{+04}\,i$
$\Delta\mu^{*,\text{off}}$	$1.000000\text{e}{+09} - 9.267354\text{e}{+04}\,i$
$\Delta\mu^{*,\text{diag}} - \Delta\mu^{*,\text{off}}$	$-3.886223\text{e}{-05} - 2.088956\text{e}{-01}\,i$

The values of T_1^c and T_2^c agree with the SymPy symbolic computation to approximately 10^{-8} relative error. T_3^c and C^c differ by approximately 0.3%, because the NumPy implementation retains higher-order Taylor terms (8 terms *vs.* the $O((\omega a)^7)$ truncation in the symbolic code).

Generated from `CubicTMatrix/effective_contrasts.py` *and* `CubicTMatrix/voigt_tmatrix.py` *using NumPy hybrid computation.* Γ_0 *evaluated by 3D Gauss–Legendre quadrature (*32^3 *points);* A^c*,* B^c*,* C^c *evaluated by polynomial Taylor expansion (8 terms) with GL quadrature.*

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